

Multiple Choice (10 marks)

1. Which of the following compounds has a ^1H resonance approximately 1.55 kHz away from TMS on a spectrometer with a 12.0 T magnet?
 - (a) CH_3F
 - (b) CH_3Cl
 - (c) CH_3Br
 - (d) CH_3I
2. Calculate the Q-factor for an X-band EPR cavity with a resonator bandwidth of 1.58 MHz.
 - (a) $Q = 1012$
 - (b) $Q = 2012$
 - (c) $Q = 3012$
 - (d) $Q = 6012$
3. Which of the following is required for both paramagnetism and ferromagnetism?
 - (a) Strong oxidizing conditions
 - (b) Low-spin electron configuration
 - (c) Metallic physical properties
 - (d) Unpaired electrons
4. Which nuclide has an NMR frequency of 115.5 MHz in a 20.0 T magnetic field?
 - (a) ^{17}O
 - (b) ^{19}F
 - (c) ^{29}Si
 - (d) ^{31}P
5. Cobalt-60 is used in the radiation therapy of cancer and can be produced by bombardment of cobalt-59 with which of the following?
 - (a) Neutrons
 - (b) Alpha particles
 - (c) Beta particles
 - (d) X-rays

True / False (10 marks)

Answer True or False

6. True or False: T_1 , unlike T_2 , is sensitive to molecular motions at the Larmor frequency.
7. True or False: The total entropy of the system plus surroundings increases during a spontaneous process.
8. True or False: Planck's quantum theory explains the emission spectra of diatomic molecules in terms of continuous energy distribution.
9. True or False: The hydrogen atom in CHCl_3 does not undergo rapid intermolecular exchange, resulting in multiple peaks in its ^1H NMR spectrum.
10. True or False: The limiting high-temperature molar heat capacity at constant volume (C_V) of a gas-phase diatomic molecule is $2.5 R$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the factors influencing the thermal stability of hydrides in group-14 elements and analyze their stability order, providing a detailed explanation of each compound's characteristics.
12. Discuss the significance of electromagnetic radiation absorption by proteins at a wavelength of 190 nm, particularly focusing on its implications in the ultraviolet region of the spectrum.

End of Paper